Assignment-1.pdf

Assignment-1.pdf is an MBAN5550 SQL and database assignment focused on using Microsoft Visual Studio and SQL Server to model and query a database for heating units, houses, technicians, and services.[1]

Assignment Structure

- The objective is to familiarize students with SQL and SQL Server tools.[1]
- The ER diagram relates houses, heating units, technicians, and their services.[1]

Questions Overview

- Create SQL tables for HOUSE, HEATINGUNIT, TECHNICIAN, and SERVICE, using appropriate primary and foreign keys.[1]
- Write and verify SQL statements for:
  - Finding the address of a house owned by John Smith.[1]
  - Listing heating unit details for 903 Robie Street.[1]
  - Counting gas and solar heating units.[1]
  - Counting gas heating units in 903 Robie Street.[1]
  - Listing unique technician titles and names servicing 903 Robie Street in a specific date range.[1]
  - Finding the house address with the largest gas heating unit.[1]
- Submission is a PDF containing all SQL statements, a cover page with name and student number, to be submitted via Brightspace.[1]

Key Tables

- HOUSE: HouseAddress, HouseOwner, Insurance.[1]
- HEATINGUNIT: HeatingUnitID, UnitName, UnitType, Manufacturer, DateOfBuilt, Capacity, HouseAddress.[1]
- TECHNICIAN: EmployeeNumber, EmployeeName, Title, YearHired.[1]
- SERVICE: HeatingUnitID, EmployeeNumber, ServiceType, Date, Time.[1]

This assignment requires applying SQL skills for table creation and database querying, and instructs verifying results in Visual Studio.[1]

write the SQL statement for creating the following tables (primary key attributes are denoted*, and foreign keys are in italic)

Here are the SQL statements for creating the tables HOUSE, HEATINGUNIT, TECHNICIAN, and SERVICE, using primary keys and foreign keys as indicated.[2]

SQL Table Definitions

```
CREATE TABLE HOUSE (
    HouseAddress VARCHAR(100) PRIMARY KEY,
    HouseOwner VARCHAR(100),
    Insurance VARCHAR(50)
);

CREATE TABLE HEATINGUNIT (
    HeatingUnitID INT PRIMARY KEY,
    UnitName VARCHAR(100),
    UnitType VARCHAR(50),
    Manufacturer VARCHAR(100),
    DateOfBuilt DATE,
    Capacity INT,
    HouseAddress VARCHAR(100),
    FOREIGN KEY (HouseAddress) REFERENCES HOUSE(HouseAddress)
);

CREATE TABLE TECHNICIAN (
    EmployeeNumber INT PRIMARY KEY,
    EmployeeName VARCHAR(100),
    Title VARCHAR(50),
    YearHired INT
);

CREATE TABLE SERVICE (
    HeatingUnitID INT,
    EmployeeNumber INT,
    ServiceType VARCHAR(50),
    Date DATE,
    Time TIME,
    PRIMARY KEY (HeatingUnitID, EmployeeNumber, Date, Time),
    FOREIGN KEY (HeatingUnitID) REFERENCES HEATINGUNIT(HeatingUnitID),
    FOREIGN KEY (EmployeeNumber) REFERENCES TECHNICIAN(EmployeeNumber)
);
```

These table definitions use typical data types and include all required primary and foreign key relationships.[2]

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1. Assignment-1.pdf

2. Assignment-1.pdf